

Foundations of Artificial Intelligence

Understanding the systems that enable machines to perform intelligent tasks

Artificial Intelligence (AI) is a broad field of computer science focused on creating systems that can perform tasks commonly associated with human intelligence. These tasks include reasoning, language understanding, pattern recognition, planning, and decision-making.

What AI Enables	AI systems can analyze large amounts of information, identify patterns, make predictions, and support decisions.
Why It Matters	AI provides the foundation for modern applications such as recommendation systems, virtual assistants, fraud detection, and automated content generation.

Machine Learning: Learning from Data

How computers discover patterns instead of relying only on explicit rules

Machine Learning (ML) is a major branch of AI in which algorithms learn relationships and patterns from examples. Rather than writing a rule for every possible situation, developers provide data from which a model can learn useful behavior.

Core Idea	A model receives training data, learns statistical patterns, and uses those patterns to make predictions or decisions on new data.
Common Applications	ML is widely used for classification, forecasting, recommendation, anomaly detection, and personalization.

Generative AI and Content Creation

Models that generate new text, images, code, audio, and other content

Generative AI focuses on producing new content based on learned patterns. A user provides an instruction or prompt, and the model generates an output that matches the requested task and context.

Typical Outputs	Generative models can create natural-language responses, software code, images, summaries, and structured information.
Key Consideration	Generated content should be evaluated because a model can produce fluent responses that are incomplete, inaccurate, or unsupported by reliable sources.

Large Language Models

The language-processing engines behind modern conversational AI

Large Language Models (LLMs) are neural-network models trained on very large collections of text and other language data. They learn statistical relationships between tokens and use those relationships to generate responses one token at a time.

What LLMs Do	LLMs can summarize, explain, translate, classify, extract information, generate code, and answer questions.
A Practical Limitation	An LLM does not automatically know private company information or current internal data. External context may need to be supplied at runtime.

Retrieval-Augmented Generation

Connecting language models to external and domain-specific knowledge

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. Instead of asking an LLM to answer only from its learned parameters, a RAG system first retrieves relevant information from an external knowledge source.

Basic Flow	A user asks a question, a retriever finds relevant document content, and the retrieved context is provided to the LLM before the answer is generated.
Why RAG Is Useful	RAG is especially useful for private documents, internal policies, product catalogs, manuals, and other information that may change over time.

Document Loaders

The entry point for bringing external documents into a RAG pipeline

Document loaders read data from different sources and convert it into a common document representation that downstream LangChain components can process. Different loaders are designed for different file formats and data sources.

Examples	TextLoader can read plain text files, while PDF loaders can extract content from PDF documents. Other loaders support formats such as web pages and office documents.
Output	A loader typically produces Document objects containing page content and metadata such as the source file or page number.

Working with PDF Documents

Extracting page-level content from PDF files for downstream processing

PDF documents often contain multiple pages with headings, paragraphs, tables, and other elements. A PDF loader extracts readable content and represents the result as Document objects that can be processed by the rest of a RAG pipeline.

Page-Level Representation	A multi-page PDF can produce multiple Document objects, allowing individual pages or sections to be retrieved later.
Important Metadata	Metadata can preserve information such as the source filename and page number, which is useful for tracing retrieved information back to its origin.

Embeddings and Semantic Representation

Turning text into numerical representations that capture meaning

Embeddings convert text into numerical vectors. These vectors provide a mathematical representation that allows systems to compare the semantic similarity between pieces of text.

Why Embeddings Matter	A question and a document chunk can use different words while expressing a similar idea. Embeddings help retrieval systems recognize this semantic relationship.
Example	A query such as 'How many days off do employees receive?' can be matched with a policy section discussing annual leave even when the wording is different.

Vector Databases

Storing embeddings and making semantic search efficient

A vector database stores numerical embeddings together with associated document information. It provides similarity-search capabilities that allow a RAG system to locate document chunks that are most relevant to a user's question.

Typical Workflow	Documents are split into chunks, chunks are embedded, and the resulting vectors are stored in a vector database.
During Retrieval	The user's question is embedded and compared with stored vectors. The most relevant chunks are returned as context for the language model.

Retrievers and End-to-End RAG

Finding the right context before the language model generates an answer

A retriever is the component responsible for selecting relevant documents or document chunks for a given query. It connects the user's question to the knowledge stored in the retrieval system.

End-to-End Flow	Question → Retriever → Relevant Context → Prompt → LLM → Answer.
Goal	The purpose of retrieval is to provide the model with useful, relevant evidence so that generated answers can be grounded in the available knowledge base.